Local Setup Guide: AI Image Detection Project

This document provides step-by-step instructions to run the project locally on a Windows machine.

Prerequisites

Before starting, ensure you have the following installed:

- **Python 3.13** (Do NOT use Python 3.15, as the dependencies will fail to compile). Download it from python.org.
- **Important:** During installation, check the box that says "Add Python to PATH".
- **Git** or a way to download the code (like downloading the ZIP from GitHub).

Step 1: Download the Code

1. Go to the GitHub repository.
2. Click the green **<> Code** button and select **Download ZIP**.
3. Extract the ZIP file to a folder on your computer (e.g., `D:\Projects\AI-Image-Detection`).
4. Open the extracted folder. You should see `app`, `data`, `requirements.txt`, etc.

Step 2: Open Terminal & Navigate to the Folder

1. Open **PowerShell** or **Command Prompt**.
2. Navigate to your project folder. Run:

```
cd "D:\Projects\AI-Image-Detection"
```

Step 3: Create a Virtual Environment

Because we are using Python 3.13, we need to explicitly tell Windows to use it to create the virtual environment:

```
py -3.13 -m venv venv
```

Activate the Virtual Environment:

```
.\venv\Scripts\Activate.ps1
```

> **Note:** If you get a "SecurityError: UnauthorizedAccess", you need to bypass the Windows execution policy by running this command first:

```
Set-ExecutionPolicy -ExecutionPolicy Bypass -Scope Process
```

Then try activating again.

(You will know the environment is active when you see `(venv)` at the start of your terminal prompt).

Step 4: Install Dependencies

With the virtual environment active, install the required packages. This might take a few minutes.

```
pip install -r requirements.txt
```

Step 5: Create the .env File

Your project needs API keys to work. These are not included in the GitHub download for security reasons.

1. Open **Notepad** or **VS Code**.
2. Create a new file named exactly `.env` (make sure Windows doesn't save it as `.env.txt`).
3. Place this file in the **project root folder**: the same level as the `app` folder and `requirements.txt`, **outside** the `app` folder (not inside it).

Your project folder should look like this:

```
AI-Image-Detection/
|-- app/                  <- code lives here (do NOT put .env inside)
|-- data/
|-- prompts/
|-- tests/
|-- venv/
|-- .env                  <- place the .env file HERE (project root)
|-- requirements.txt
```

> **Correct:** `AI-Image-Detection\.env` (next to the `app` folder)
> **Wrong:** `AI-Image-Detection\app\.env` (inside the `app` folder)

4. Paste your API keys inside:

```
GEMINI_API_KEY=your_actual_gemini_api_key_here
GEMINI_MODEL=gemini-3.5-flash-lite
GEMINI_FALLBACK_MODELS=gemini-2.5-flash-lite
GEMINI_MAX_IMAGE_SIZE=1280
GEMINI_JPEG_QUALITY=90
SIGHTENGINE_API_USER_1=your_actual_sightengine_user
SIGHTENGINE_API_SECRET_1=your_actual_sightengine_secret
SIGHTENGINE_API_USER_2=
SIGHTENGINE_API_SECRET_2=
SIGHTENGINE_AI_THRESHOLD=0.40
OFFICE_RADIUS_METRES=100
```

Save the file.

Step 6: Run the Server

Ensure you are still in the `(venv)` environment and in the root folder of the project. Run:

```
uvicorn app.main:app --reload
```

You should see output indicating the server is running:

```
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process...
INFO:     Application startup complete.
```

Step 7: Test the Application

1. Open your web browser (Chrome, Edge, etc.).
2. Go to: `http://127.0.0.1:8000`
3. Open the **"Test setup"** section in the UI.
4. Leave the **API Base URL** blank (it will default to localhost).
5. Click **"Test connection"**. It should show green text saying Gemini and Sightengine are ready.

You can now use the UI to upload images and test the system!